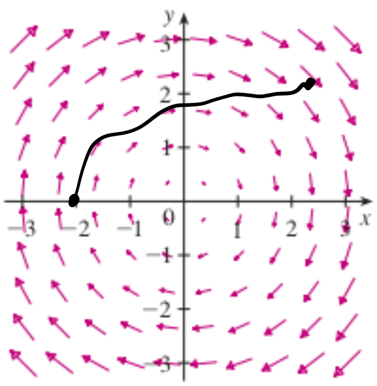
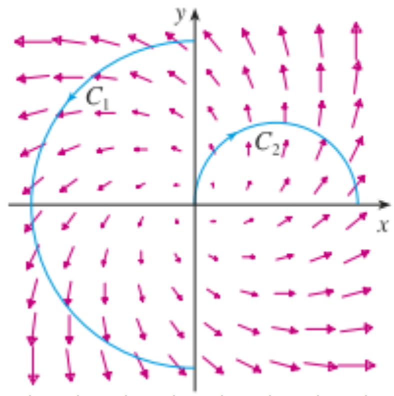


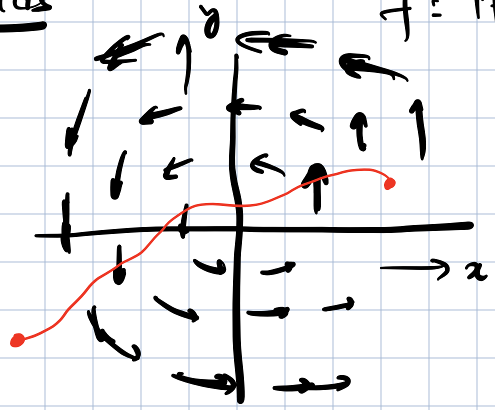
y
x
−3
−2
−1
0
1
2
3



y
x
C_1
C_2

Vector fields

$f: \mathbb{R}^2 \to \mathbb{R}^2 \qquad f: \mathbb{R}^3 \to \mathbb{R}^3.$



$$f(x,y) = -y\,\vec{i} + x\,\vec{j}$$

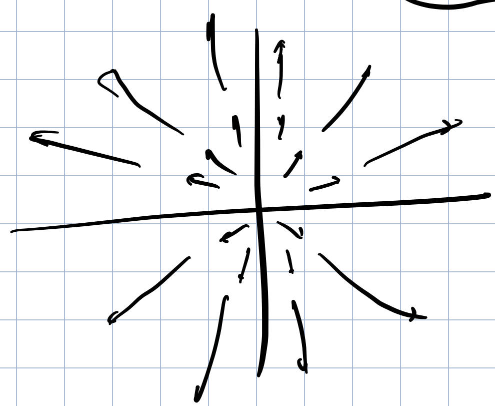
$$\begin{pmatrix} -y \\ x \end{pmatrix}$$

perp to $\begin{pmatrix} x \\ y \end{pmatrix}$

Examples: Force field.

For one-one maps, the Jacobian is an invertible matrix).

line integrals & ($\underbrace{\text{surface integrals}}$)



$$F(x) = \int_0^x f(a)\,da$$

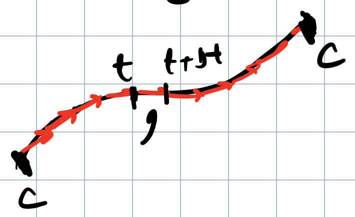
$$F'(x) = f.$$

(11-12 LHC10)

Recap: Vector fields are $f: \mathbb{R}^3 \to \mathbb{R}^3$ or $f: \mathbb{R}^2 \to \mathbb{R}^2$

For a vector field f, the line integral / path integral along a (smooth) curve C is defined as

$$\int_C \vec{f} \cdot d\vec{r} = \int_{t=a}^{t=b} f(\vec{r}(t)) \cdot \vec{r}\,'(t)\,dt$$

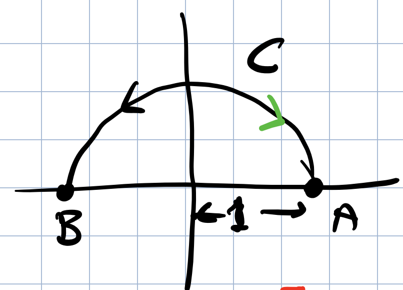


Let C be defined as

$$C = \{\vec{r}(t) \text{ for } a \le t \le b\}$$

$$d\vec{r} \approx r'(t)\,\Delta t$$

Example: Find the line integral of $-y\vec{i} + x\vec{j}$ on the semicircle of radius 1 shown.



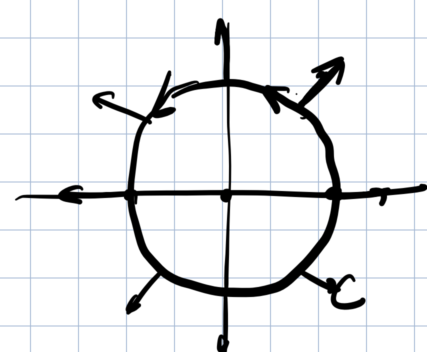
Semicircle: $r(\theta) = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$ for $0 \le \theta \le \pi$

$$r'(\theta) = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix} \qquad -\sin\theta\,\vec{i} + \cos\theta\,\vec{j}$$

$$\int_0^\pi \underbrace{\left[-\sin\theta\,\vec{i} + \cos\theta\,\vec{j}\right]}_{f(r(\theta))} \cdot \underbrace{\left[-\sin\theta\,\vec{i} + \cos\theta\,\vec{j}\right]}_{r'(\theta)} d\theta$$

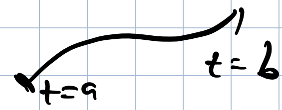
$$= \pi$$

What is the line integral on circle



Exer: line integral of $x\vec{i} + y\vec{j}$ along C on the right?

Property: Let the vector field be the gradient of a scalar valued function $f(x, y)$.



Then, $$\int_C \nabla f \cdot d\vec{r} = \int_{t=a}^{t=b} \frac{df}{dt}\, dt$$

$$r(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

$$\left(\frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j}\right) \cdot \underbrace{\left(\frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j}\right)}_{}^{r'(t)}$$

$$= \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

$$= \frac{df}{dt} \quad \text{by chain rule}$$

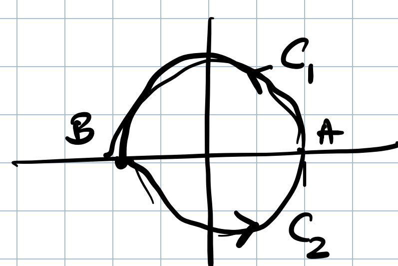
$$\boxed{\int_C \nabla f \cdot d\vec{r} = f(\underbrace{r(b)}_{\text{final position}}) - f(\underbrace{r(a)}_{\text{initial position}})}$$

FTIC $$\int_a^b f'(x)\,dx = f(b) - f(a)$$

Conclusion: 1) Generalisation of fundamental theorem of integral calculus

2) line Integral of a gradient field depends only the initial & final points in the path.

Exc:



$$\int_{C_1} \nabla f \cdot d\vec{r} = f(B) - f(A)$$

$$\int_{C_2} \nabla f \cdot d\vec{r} = f(A) - f(B)$$

$$\int_{C} \nabla f \cdot d\vec{r} = 0$$

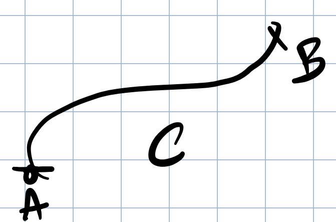
3) Line integral along a closed curve for any gradient field is zero.

Excercu If $-y\vec{i} + x\vec{j}$ a gradient field?

No as line integral along unit circle is $2\pi \neq 0$.

- $\oint_C$ indicates line integral on closed curve
- Gradient fields are conservative.

Converse also holds (provided the partial derivatives exist & are continuous).



Can show that

$$\int_C \nabla f \cdot d\vec{r} \quad \leftarrow \nabla f = m\vec{r}''(t)$$

$$= \frac{m}{2}\left[|r'(A)|^2 - |r'(B)|^2 \right]$$

$$= f(B) - f(A)$$

$$f(A) + \frac{m}{2}|r'(A)|^2 = f(B) + \frac{m}{2}|r'(B)|^2$$

How to check if a vector field is conservative / gradient field?

Approach 1: Integrate on closed paths & check if zero.

Approach 2: $P(x,y)\vec{i} + Q(x,y)\vec{j}$

$$P = \frac{\partial f}{\partial x} \qquad Q = \frac{\partial f}{\partial y}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

$P(x,y)\vec{i} + Q(x,y)\vec{j}$ is a gradient field

if & only if $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$. (provided P, Q are continuous & partial derivatives exist).

Example: $\underbrace{-y}_{P}\vec{i} + \underbrace{x}_{Q}\vec{j}$

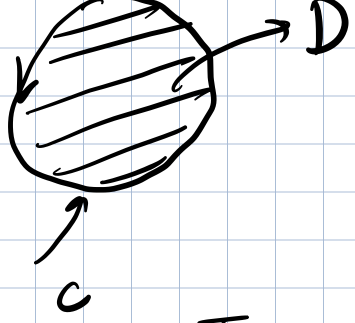
$$\frac{\partial Q}{\partial x} = 1 \qquad \frac{\partial P}{\partial y} = -1$$

Green's theorem Let $F = P\vec{i} + Q\vec{j}$ be a vector field in $\mathbb{R}^2$.

(Version 1) $$\oint_C \vec{F}\cdot d\vec{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

assuming counterclockwise orientation ∂C.

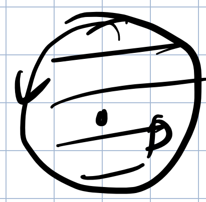
$\int^b f'(x)\,dx$



$$\int_a \; = f(b) - f(a)$$

Example: $\vec{F} = -y\vec{i} + x\vec{j}$ $\quad C: x^2 + y^2 = 1$

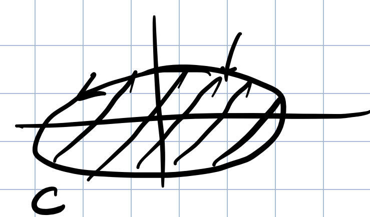
$$LHS = \oint_C \vec{F}\cdot d\vec{r} = 2\pi \text{ (computed earlier)}$$



$$\frac{\partial Q}{\partial x} = 1, \quad \frac{\partial P}{\partial y} = -1$$

$$RHS = \iint_D 2\,dA = 2\,\text{area}(D) = 2\pi.$$

Example: Find area of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ using only single integral.



$$\vec{F} = -y\vec{i} + x\vec{j}$$

$$\oint_C \vec{F}\cdot d\vec{r} = 2\iint_D dA$$

$$r(\theta) = \begin{pmatrix} a\cos\theta \\ b\sin\theta \end{pmatrix} \qquad 0 \le \theta \le 2\pi$$

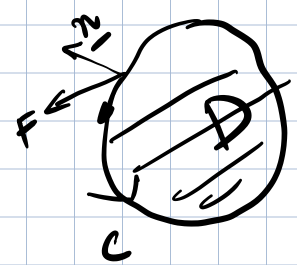
$$r'(\theta) = \begin{pmatrix} -a\sin\theta \\ b\cos\theta \end{pmatrix}$$

2π

$$\int_0 \left(\cancel{a} b \sin\theta\,\vec{i} + a\cos\theta\,\vec{j} \right) \cdot \left(-a\sin\theta\,\vec{i} + b\cos\theta\,\vec{j} \right) d\theta$$

$$= 2\pi a b.$$

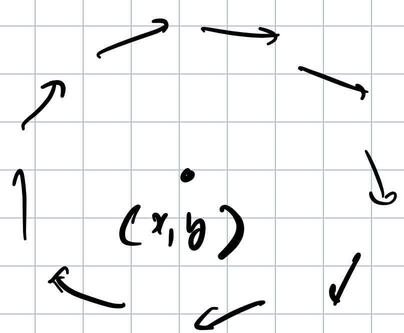
<u>(Version 2)</u> : Let $\vec{F} = P\vec{i} + Q\vec{j}$ be a vector field.



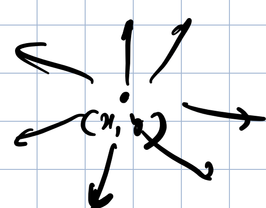
$$\int_C (\vec{F}\cdot n)\, dr = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) \partial A$$

$\dfrac{\partial P}{\partial x} + \dfrac{\partial Q}{\partial y}$:

$\dfrac{\partial Q}{\partial x} - \dfrac{\partial P}{\partial y}$: circulation locally around (x, y)



$\dfrac{\partial P}{\partial x} + \dfrac{\partial Q}{\partial y}$: net outward flow locally around x, y



Curl of a vector field:

$\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$

$$\operatorname{curl} F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

written $\vec{\nabla} \times \vec{F}$.

$$= \vec{i}\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) - \vec{j}\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + \vec{k}\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$$

$F = -y\vec{i} + x\vec{j}$

$\operatorname{curl} F = 2\vec{k}$

curl F : directed towards the local "axes of rotation"

magnitude is high for "faster" rotation.

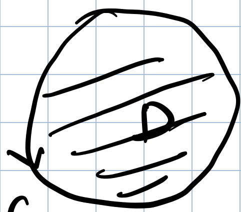
Property: $\operatorname{curl}(\nabla f) = 0$.

Exercise: Proof.

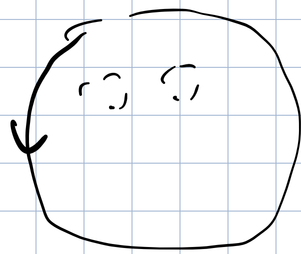
A vector field F is conservative (gradient field) if & only if $\operatorname{curl}(F) = 0$ throughout.

Stokes' theorem: Generalization of Green's theorem to 3 dimensions

$$\iint_D (\text{curl } F \cdot n)\, dA = \oint_C \vec{F} \cdot d\vec{r}$$



C



Divergence: For any vector field $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$,

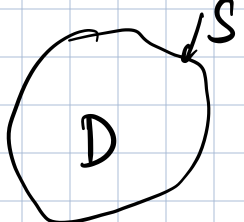
the divergence is defined as

$$\text{div } F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = \nabla \cdot F$$

$$\nabla \to \frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} \; \frac{\partial}{\partial z}\vec{k}$$

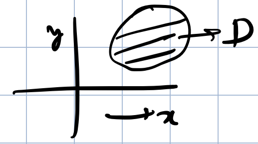
Divergence theorem:

$$\iiint_D (\text{div } F)\, dV = \underbrace{\int_S (\vec{F} \cdot \vec{n})\, dA}_{\text{Flux:}}$$

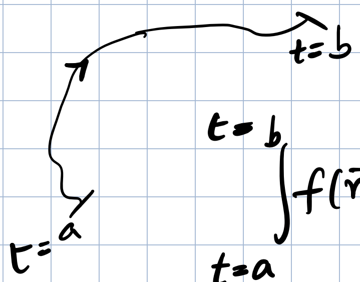


1) Area & volume integrals $\iint_D f(x,y)\,dA$

(Scalar valued fn)

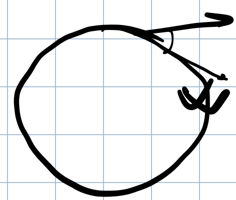


2) Line & surface integrals



$\vec{r}(t)$

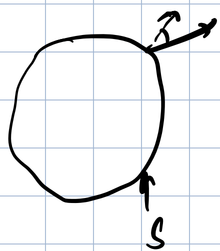
$$\int_{t=a}^{t=b} f(\vec{r}(t)) \cdot \vec{r}\,'(t)\,dt$$



$$\nabla \times \underline{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

Surface integral:

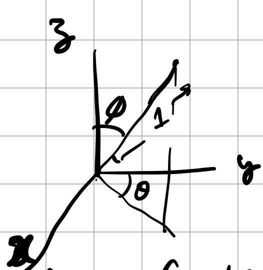
Flux



$$\int_S \vec{F} \cdot \vec{n}\,ds$$

Surface is parameterized by u, v : $\vec{r}(u,v)$ gives all points on the surface, for various u, v.

Example: parameterizing a sphere:



$$\vec{r}_\theta = \begin{pmatrix} \sin\phi\cos\theta \\ -\sin\phi\sin\theta \\ 0 \end{pmatrix}, \quad \vec{r}_\phi = \begin{pmatrix} \cos\phi\sin\theta \\ \cos\phi\cos\theta \\ -\sin\phi \end{pmatrix}$$

$$\vec{r}(\theta,\phi) = (\sin\phi\sin\theta,\ \sin\phi\cos\theta,\ \cos\phi)$$

$0 \le \theta \le 2\pi$

$0 \le \phi \le \pi$

tangent direction when u is changed:

$$\vec{r}(u+\Delta u, v) - \vec{r}(u,v) = \begin{bmatrix} \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial u} \\ \frac{\partial z}{\partial u} \end{bmatrix} \Delta u$$

Let $\vec{r}_u = \begin{bmatrix} \partial x/\partial u \\ \partial y/\partial u \\ \partial z/\partial u \end{bmatrix}$, $\vec{r}_v = \begin{bmatrix} \partial x/\partial v \\ \partial y/\partial v \\ \partial z/\partial v \end{bmatrix}$

Then the surface integral is computed as

$$\iint_{u,v} f(\vec{r}(u,v)) \cdot (\vec{r}_u \times \vec{r}_v)\, dA .$$

divergence : 1

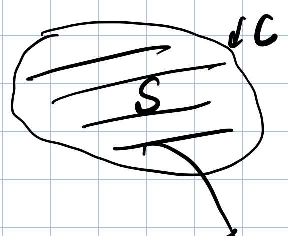
Exercise: Find the flux of the field $z\vec{i} + y\vec{j} + x\vec{k}$ through the unit sphere centered at the origin,

Compute $r_\theta \times r_\phi$, and verify it points outward.

Compute dot product & evaluate integral.

↳ compare with divergence thm: $\frac{4\pi}{3}$

Stokes theorem



Surface "without holes"

1) $$\iint_S \operatorname{curl} F \cdot \vec{n}\, ds = \oint_C \vec{F} \cdot d\vec{r}$$

2) Green's theorem,

2-d vector field $f: \mathbb{R}^2 \to \mathbb{R}^2$ $F = P\vec{i} + Q\vec{j}$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$\operatorname{div} F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

3) Divergence thm:

$$\oiint_{\text{closed surface}} \vec{F} \cdot \vec{n}\, ds = \iiint (\operatorname{div} F)\, dV .$$

Exercise: What is the flux of curl F through a closed surface. Ans: 0

Property: div curl $F = 0$ for any F.

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$\left(\cancel{\frac{\partial^2 R}{\partial x \partial y}} - \cancel{\frac{\partial^2 Q}{\partial x \partial z}}\right) + \left(\cancel{\frac{\partial^2 P}{\partial y \partial z}} - \cancel{\frac{\partial^2 R}{\partial y \partial x}}\right) + \left(\cancel{\frac{\partial^2 Q}{\partial z \partial x}} - \cancel{\frac{\partial^2 P}{\partial z \partial y}}\right) = 0.$$

Exercise: $$f(x,y,z) = \frac{-1\left(x\vec{i} + y\vec{j} + z\vec{k}\right)}{\left(x^2+y^2+z^2\right)^{3/2}}$$

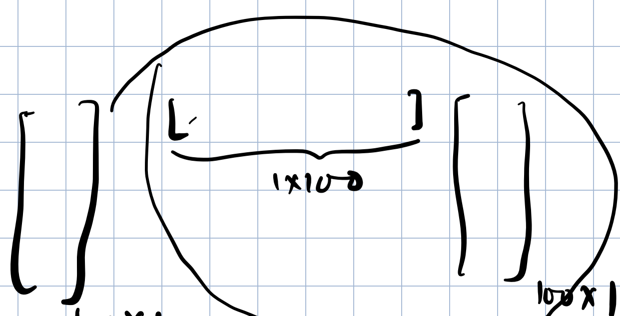
Compute divergence & curl: $\rightarrow$ should also be zero everywhere in domain.
Should be equal to 0 everywhere except $x=0, y=0, z=0$.

Topics for Final exam

1) Linear algebra: (in)dependence, one-one & onto functions, rank, nullity, computational aspects (multiplying matrix with matrix)

(QR decomposition
solve least squares problems)

Geometry (angles, distances)



1a) General vector spaces.

2) Sequences/series: monotone convergence theorem, squeeze thm, tests for convergence of series, approximating series with a few terms, Taylor series

3) Gradient & Jacobian

4) $\underbrace{\text{Integrals}}_{\text{2-D/3-D}}$, Change of variable in integrals.

5) Greens, stokes & divergence theorems. }